

Empirical Lévy Scaling in Short-Horizon Futures Returns: Estimation, Sensitivity, and Autocorrelation Bias

Haoping Qiu
[Department]
[University]
[City, Country]
[email]

Abstract—Short-horizon price increments in equity-index futures exhibit heavy tails and a sharply peaked central body inconsistent with Gaussian dynamics, motivating Lévy-stable distributional models with characteristic exponent $\alpha \in (1, 2)$. This paper examines Lévy scaling across eight equity-index futures contracts spanning North America and Europe using 1-minute OHLCV data from exchange inception through January 2026. Three estimation approaches — peak-scaling OLS, collapse-quality minimisation, and fractional-order structure functions — are applied across a full-factorial robustness matrix spanning histogram resolution, lag-grid density, and intraday seasonality correction. Seven of the eight markets yield peak-scaling exponents $\hat{\alpha}_{\text{tip}} \in (1.73, 1.84)$ with $R^2 > 0.99$, consistent with the truncated Lévy-flight consensus. The S&P/TSX 60 contract is a persistent outlier ($\hat{\alpha} \approx 2.6$), attributable to low liquidity and excess zero increments. The estimator choice, not tuning parameters, is the dominant source of variation in reported exponents; inter-estimator spreads of 0.15–0.35 units exceed the inter-market range for the seven conforming contracts. Seven of the eight markets exhibit negative first-order autocorrelation and one (ER) a marginally positive value, all violating the i.i.d. assumption underlying peak-scaling. A closed-form variance-ratio bias correction reduces $\hat{\alpha}$ by 0.04–0.92 units for seven of the eight markets; the sole exception (ER) has marginally positive first-order autocorrelation and corrects upward by 0.04 units. The correction magnitude is monotonically ordered by variance ratio across the seven mean-reverting markets, providing a cross-sectional validation of the bias mechanism.

Index Terms—Lévy-stable distributions, peak-scaling estimator, equity-index futures, variance ratio, intraday returns, self-similarity.

I. INTRODUCTION

The distributional properties of short-horizon price changes have been a central concern of financial economics and econophysics since at least Bachelier [1]. The Gaussian random-walk model, which predicts normally distributed increments and variance scaling linearly with the holding period, is a natural first approximation but a poor empirical description of intraday returns. High-frequency data reveal distributions with heavy tails, excess kurtosis, and a central body that is far more peaked than the Gaussian. These features, documented across equities, foreign exchange, and derivatives markets, motivate the use of Lévy-stable distributions as a richer distributional model [2]–[4].

The central prediction of the symmetric α -stable model is *self-similarity*: the distribution of τ -step price increments is a rescaled version of the one-step distribution, with the scale factor growing as $\tau^{1/\alpha}$ for a characteristic exponent $\alpha \in (1, 2)$. This implies, in particular, that the peak of the empirical probability density decays as $P(0, \tau) \propto \tau^{-1/\alpha}$, a relation that can be estimated by ordinary least squares on a log-log plot and that forms the basis of the *peak-scaling* approach of Mantegna and Stanley [3], [5]. The same self-similarity predicts that empirical PDFs at different lags, when rescaled by $\tau^{1/\alpha}$, should collapse onto a common curve, providing a qualitative visual test of the Lévy hypothesis. Both the regression and the collapse have been applied extensively in the econophysics literature, with estimated exponents consistently falling in the range $\alpha \approx 1.5$ – 1.9 [6]–[8].

Despite this activity, three important gaps remain. First, most existing studies focus on a single market or a narrow class of instruments, typically US equity indices or individual stocks, and it is unclear how far the $\alpha \approx 1.7$ – 1.8 consensus extends to a focused cross-section of eight liquid equity-index futures contracts across different geographies and regulatory environments. Second, the sensitivity of $\hat{\alpha}$ to methodological choices including histogram resolution, lag grid density, intraday seasonality removal, and estimator selection has not been systematically quantified in a single study; different choices can shift estimates by amounts that matter for practical inference. Third, and most substantively, the peak-scaling estimator is derived under the assumption of independent increments, yet 1-minute futures returns exhibit well-documented short-term mean reversion driven by the bid-ask bounce and order-book dynamics [9], [10]. The direction of the resulting bias is known qualitatively [8], but no closed-form correction has been derived and validated empirically.

This paper addresses all three gaps. Using 1-minute OHLCV data for eight equity-index futures contracts spanning North America and Europe, sourced from OneTick via TickData and covering each contract from exchange inception through January 2026, we make six contributions organized around the three gaps.

Addressing gap 1 (cross-market coverage): (i) We establish

a baseline peak-scaling survey across all eight markets, reporting $\hat{\alpha}_{\text{tip}}$ and collapse diagnostics for each. (v) We test the symmetry assumption of the Lévy model by computing the pointwise asymmetry of the empirical PDFs and examining whether symmetrization improves collapse quality.

Addressing gap 2 (methodological sensitivity): (ii) We test the effect of intraday seasonality removal on $\hat{\alpha}$, running the full PDF analysis on both raw and de-seasonalized price series and comparing the resulting estimates. (iii) We compare three independent estimators — tip-scaling OLS, collapse-quality minimization, and fractional-order structure functions — across all markets to assess their mutual consistency. (iv) We construct a full-factorial robustness matrix spanning all combinations of histogram resolution, lag-grid density, preprocessing choice, and estimator, and decompose the variance of $\hat{\alpha}$ across these factors to identify which choices matter most.

Addressing gap 3 (bias correction): (vi) We derive a closed-form bias correction for the peak-scaling estimator under short-term mean reversion, express it in terms of the Campbell–Lo–MacKinlay variance ratio, and validate the correction across all eight markets.

The paper is organized as follows. Section II provides the theoretical background on Lévy-stable distributions, peak-scaling estimation, and the variance-ratio framework. Section III describes the data and preprocessing steps. Section IV develops the six analytical methods in detail. Section V presents the results for each contribution. Section VI summarizes the findings, discusses limitations, and suggests extensions.

II. BACKGROUND

A. Lévy-stable distributions and the peak-scaling formula

The Lévy-Khintchine theorem establishes that the class of stable distributions is the complete set of non-trivial distributional limits of normalized sums of i.i.d. random variables [11], [12]. A symmetric α -stable variable $X \sim S_\alpha(\gamma, 0, 0)$ has characteristic function

$$\mathbb{E}[e^{iqX}] = \exp(-\gamma|q|^\alpha), \quad \gamma > 0, \quad \alpha \in (0, 2]. \quad (1)$$

The Gaussian ($\alpha = 2$) is the unique finite-variance special case; for $\alpha < 2$ the tails decay algebraically as $P(|X| > u) \sim C_\alpha u^{-\alpha}$, so all moments of order $\geq \alpha$ are infinite.

For i.i.d. increments drawn from $S_\alpha(\gamma_1, 0, 0)$, the τ -step sum has scale $\gamma = \gamma_1^\alpha \tau$, yielding the self-similarity property

$$P_{\alpha, \gamma}(x, \tau) = (\gamma\tau)^{-1/\alpha} F_\alpha\left(\frac{x}{(\gamma\tau)^{1/\alpha}}\right), \quad (2)$$

where F_α is the standardized stable density. This scaling form is the central prediction of the Lévy model: the shape of the increment distribution is identical at every lag, up to a scale factor growing as $\tau^{1/\alpha}$. Setting $x = 0$ and using $F_\alpha(0) = \Gamma(1/\alpha)/(\pi\alpha)$ gives the *peak-scaling formula*

$$P(0, \tau) = \frac{\Gamma(1/\alpha)}{\pi\alpha} (\gamma\tau)^{-1/\alpha}. \quad (3)$$

The log-log regression of $\hat{P}(0, \tau)$ on τ therefore has slope $-1/\alpha$, independent of the scale parameter γ .

Mandelbrot [2] first proposed that asset-price increments lie in the domain of attraction of a stable law with $\alpha \in (1, 2)$. Mantegna and Stanley [3], [5] provided the first systematic intraday evidence using S&P 500 data, finding $\alpha \approx 1.4$ over three orders of magnitude in lag and demonstrating clean PDF collapse. The truncated Lévy flight (TLF) resolves the tension between infinite-variance stable laws and finite empirical moments: the distribution follows the stable power law over intermediate scales but is exponentially cut off in the far tail, producing finite variance while preserving Lévy-like body scaling over the observable range [3], [8]. Weron [7] showed that samples of 10^6 or more are required before the true stable tail emerges, far beyond the practical size of intraday datasets, so empirical $\hat{\alpha}$ estimates are best interpreted as body-scaling parameters rather than tail-index claims.

Gopikrishnan et al. [6] extended the analysis to individual US stocks and found power-law tails with exponent $\zeta \approx 3$, outside the Lévy-stable regime in the strict sense. Weron [7] showed this apparent discrepancy is a finite-sample artifact: a Lévy-stable distribution with $\alpha \approx 1.7$ - 1.8 produces Hill-estimator values of $\zeta \approx 3$ on samples of realistic size. This reconciliation motivates the multi-estimator approach of the present paper and the conservative interpretation of $\hat{\alpha}$ as a central-body parameter. More recent work by Masoliver and Perelló [13] and Cont et al. [14] confirms that the central-body Lévy-scaling picture is stable through the 2010s, justifying its application to contemporary futures data.

B. Estimation of the scaling exponent

Three estimation strategies are used in this paper.

The *peak-scaling OLS* estimator, introduced by Mantegna and Stanley [3], regresses $\log \hat{P}(0, \tau)$ on $\log \tau$ and recovers $\hat{\alpha}_{\text{tip}} = -1/\hat{m}$ from the slope $\hat{m}$. The estimator is semi-parametric: it uses only the value of the distribution at the origin and makes no assumption about the shape elsewhere.

The *collapse-quality* estimator [15] chooses α to minimize the spread of the rescaled empirical curves $\tau^{1/\alpha} \hat{P}(\tau^{1/\alpha} \tilde{x}, \tau)$ around a common function. A mean-squared log-deviation score weights the central body and the tails more evenly than a pure-origin regression, providing an independent route to $\hat{\alpha}$ that does not rely on a single histogram bin.

The *fractional structure function* estimator [8], [16] exploits the prediction $S_q(\tau) = \mathbb{E}[|\Delta P_\tau|^q] \propto \tau^{q/\alpha}$ for $q < \alpha$. A log-log regression of $\hat{S}_q(\tau)$ on τ yields slope $q/\hat{\alpha}_q$; the median across moment orders $q \in \{0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$ gives $\hat{\alpha}_{\text{sf}}$. Fractional orders $q \in (0, 1)$ downweight outliers and converge reliably on moderate samples [17], while $q \in (1, \alpha)$ are sensitive to the tail. Systematic concavity of the empirical scaling exponent $\hat{\zeta}(q) = q/\hat{\alpha}_q$ would signal multifractal corrections beyond simple Lévy scaling [16], [18].

Weron [7] and Baručík and Vařha [19] document that all three of these semi-parametric methods outperform the Hill estimator for α near 2 on finite samples. The comparison of

the three estimators across all eight markets is central to the robustness analysis of this paper.

C. Short-term dependence and the variance-ratio bias

Lo and MacKinlay [20] introduced the variance ratio

$$VR(\tau) = \frac{\text{Var}(\Delta P_\tau)}{\tau \text{Var}(\delta_1)}, \quad (4)$$

which equals 1 under i.i.d. increments, exceeds 1 under positive serial correlation, and falls below 1 under mean reversion. The exact relationship to the autocorrelation function is

$$VR(\tau) = 1 + 2 \sum_{k=1}^{\tau-1} \left(1 - \frac{k}{\tau}\right) \rho(k), \quad (5)$$

which forms the basis for both variance-ratio estimators implemented in the notebook.

At 1-minute frequency the dominant effect is mean reversion, not the positive correlation found at weekly horizons by Lo and MacKinlay [20]. Two mechanisms contribute. Roll [9] showed that bid-ask bounce generates spurious first-order autocorrelation $\hat{\rho}(1) \approx -s^2/4$ for effective spread s , a purely mechanical effect. Bouchaud et al. [10] and Farmer et al. [21] documented that large market orders move prices but this impact partially reverts as the order book replenishes, adding genuine mean reversion at short horizons. Bouchaud and Potters [8] (Ch. 6) report $\hat{\rho}(1) \approx -0.1$ to -0.2 at 1-minute frequency in major equity index futures, consistent with the values found in this paper.

The canonical peak formula (3) is derived under strictly independent increments. When $VR(\tau) \neq 1$, the effective Lévy scale at lag τ is modified. The derivation proceeds in two steps. First, by the definition of the variance ratio in Equation (4), the spread of ΔP_τ under correlated increments satisfies $\text{Var}(\Delta P_\tau) = \tau \text{Var}(\delta_1) \cdot VR(\tau)$, i.e., the “dispersion” of the τ -step sum is scaled by $\sqrt{VR(\tau)}$ relative to the i.i.d. baseline. Second, for a symmetric α -stable distribution the scale parameter γ transforms as $\gamma \mapsto c^\alpha \gamma$ under a linear scaling $X \mapsto cX$ (homogeneity of the stable CF); hence scaling the dispersion by $\sqrt{VR(\tau)}$ scales γ by $(\sqrt{VR(\tau)})^\alpha = VR(\tau)^{\alpha/2}$. Combining, the effective Lévy scale at lag τ is

$$\gamma_{\text{eff}}(\tau) = \gamma \tau \cdot VR(\tau)^{\alpha/2}, \quad (6)$$

and substituting into Equation (3) gives the corrected log-peak regression

$$\log P(0, \tau) = -\frac{1}{\alpha} \log \tau - \frac{1}{2} \log VR(\tau) + \text{const.} \quad (7)$$

For a mean-reverting process ($VR(\tau) < 1$) the omission of the second term causes the raw OLS slope to underestimate $-1/\alpha$ in magnitude, biasing $\hat{\alpha}_{\text{tip}}$ upward toward the Gaussian boundary. The correction procedure is developed in Section IV-G. To our knowledge, the explicit OLS bias formula in the Campbell–Lo–MacKinlay variance-ratio form and its cross-market empirical validation have not been jointly reported; Bouchaud and Potters [8] (Ch. 6) discuss the qualitative direction of the effect but do not derive the specific correction term or validate it across a cross-section of markets.

III. DATA

A. Markets and sample

The dataset consists of 1-minute resolution day-session price histories for eight equity-index futures contracts spanning four geographical regions. The data were provided by Prof. Alexei Chekhlov and sourced from OneTick via Tick-Data, covering each contract from its exchange inception through 14 January 2026. Table I provides a full description of each contract.

The eight contracts were selected to provide broad coverage across asset classes and geographies rather than on the basis of any statistical pre-screening. Three North American equity-index contracts (YM, NQ, ER, ES) trade on CME Group venues; one Canadian contract (PT) trades on the Montreal Stock Exchange; and three European contracts (FT, DA, CF) cover the major continental and UK equity benchmarks.

The raw files are stored in “skip” format: minutes with no transactions are omitted from the file rather than recorded as zero-volume observations. This is addressed in preprocessing (Section III-B).

To justify the use of 1-minute price changes as the elementary increment, Table II reports the standard deviation of 1-minute price changes σ_1 and the average bid-ask spread Δ for each market. A necessary condition for 1-minute returns to carry genuine price information is $\sigma_1/\Delta > 1$; otherwise a substantial fraction of observed changes reflects the bid-ask bounce rather than true price movement [9]. As the ratio σ_1/Δ exceeds 1 for all eight markets, the 1-minute frequency is appropriate.

B. Preprocessing

Three preprocessing steps are applied uniformly across all markets before any statistical analysis.

Integer-scale conversion. Raw prices are quoted in their native tick units. To work in integer arithmetic and avoid floating-point accumulation errors, each price series is multiplied by the contract-specific scaling factor $1/\Delta$ and rounded to the nearest integer, so that one tick corresponds to one integer unit. All subsequent price differences are therefore integer-valued.

Gap-filling. The raw files omit minutes with no transactions. Missing minutes within a trading session are filled by carrying forward the last observed price (last-observation-carried-forward, LOCF). This preserves the equal-spacing of the 1-minute grid required for the lag- τ increment construction while recording zero price change during illiquid periods, which is consistent with the true mid-price behavior in the absence of transactions.

Lag grid construction. Price increments $\delta_\tau(t) = P(t) - P(t - \tau)$ are formed for each lag τ on two log-spaced grids. The coarse grid ($N_\tau = 20$) spans $\tau \in \{1, 2, 3, 6, \dots, 56,234\}$ minutes; the fine grid ($N_\tau = 37$) spans the same range with higher density. The maximum lag of 56,234 minutes corresponds to approximately six months of trading for a 405-minute session, well beyond the scale at which convergence to Gaussian behavior is expected. Both grids are used in all estimations to assess the sensitivity of $\hat{\alpha}$ to the choice of N_τ .

TABLE I
SAMPLE FUTURES CONTRACTS

Symbol	Name	Region	Exchange	History Start	Session
YM	Dow Jones E-mini	Americas	CBOT/CME	May 2002	08:30–15:15
NQ	NASDAQ 100 E-mini	Americas	CME	Jul 1999	08:30–15:15
ER	Russell 2000 E-mini	Americas	CBOT/CME	Jan 2002	08:30–15:15
ES	S&P 500 E-mini	Americas	CME	Sep 1997	08:30–15:15
PT	S&P Canada 60	Americas	Montreal SE	Oct 1999	09:30–16:15
FT	FTSE 100	Europe	ICE Futures	Jul 1998	08:00–16:30
DA	DAX	Europe	EUREX	Jan 1997	09:00–17:30
CF	CAC-40	Europe	Euronext	May 2000	09:00–17:30

TABLE II
MARKET VOLATILITY AND BID-ASK SPREAD

Market	σ_1	Δ (bid-ask)	σ_1/Δ
YM	6.15	1.30	4.73
NQ	2.33	0.48	4.90
ER	0.51	0.22	2.32
ES	0.68	0.25	2.71
PT	2.78	2.50	1.11
FT	3.48	1.20	2.90
DA	2.61	2.00	1.31
CF	0.46	0.22	2.09

IV. METHODOLOGY

A. Empirical PDF estimation

For each market and lag τ on the grid, price increments $\Delta P_\tau(t) = P(t) - P(t - \tau)$ are formed from the gap-filled integer-scaled price series. The histogram bin width is $\Delta x = \varepsilon \cdot \Delta$, where Δ is the bid-ask spread in tick units and ε is a dimensionless resolution parameter. The normalised histogram density is

$$\hat{P}(x_j, \tau) = \frac{\#\{i : \Delta P_\tau(t_i) \in \text{bin } j\}}{N_\tau \cdot \Delta x}, \quad (8)$$

where N_τ is the number of available increments. The bin half-width $x_{\max}$ is set to cover the bulk of the distribution while maintaining $N_x = 2,000$ bins. The empirical peak $\hat{P}(0, \tau)$ is the density of the bin whose centre is nearest $x = 0$.

The resolution parameter ε is treated as a sensitivity dimension: all analyses are run at $\varepsilon \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ and the sensitivity of $\hat{\alpha}$ to this choice is quantified in the robustness matrix (Section IV-F).

B. Baseline peak-scaling estimator

Taking logs of Equation (3) gives the tip regression

$$\log \hat{P}(0, \tau) = -\frac{1}{\alpha} \log \tau + c + u_\tau, \quad (9)$$

where $c = \log[\Gamma(1/\alpha)/(\pi\alpha)] - \log(\gamma)/\alpha$ is a nuisance intercept and u_τ is a residual. Ordinary least squares on Equation (9) over the full τ grid yields the slope $\hat{m}$ and the baseline estimate $\hat{\alpha}_{\text{tip}} = -1/\hat{m}$. The coefficient of determination R^2 is reported as a diagnostic for the power-law linearity assumption; a high R^2 is necessary but not sufficient for the Lévy model.

C. Collapse-quality estimator

For a trial exponent α , define the rescaled coordinate $\tilde{x} = x/\tau^{1/\alpha}$ and the rescaled density $\tilde{P}(\tilde{x}, \tau) = \tau^{1/\alpha} \hat{P}(x, \tau)$. Under exact Lévy self-similarity all rescaled curves collapse onto the common function F_α . Collapse quality is measured by

$$\text{Score}(\alpha) = \frac{1}{|\mathcal{G}||\mathcal{T}|} \sum_{\tilde{x}_j \in \mathcal{G}} \sum_{\tau \in \mathcal{T}} [\log \tilde{P}(\tilde{x}_j, \tau) - \mu_j]^2, \quad (10)$$

where $\mu_j = |\mathcal{T}|^{-1} \sum_{\tau} \log \tilde{P}(\tilde{x}_j, \tau)$ is the cross- τ log-mean at grid point $\tilde{x}_j$ and $\mathcal{G}$ is a common evaluation grid in $\tilde{x}$. The collapse estimator $\hat{\alpha}_{\text{collapse}}$ is the minimiser of $\text{Score}(\alpha)$ over a grid of 160 candidate values in $[1.2, 2.8]$.

D. Structure-function estimator

For each moment order $q \in \{0.25, 0.50, 0.75, 1.00, 1.25, 1.50\}$, the empirical structure function is

$$\hat{S}_q(\tau) = \frac{1}{N_\tau} \sum_t |\Delta P_\tau(t)|^q. \quad (11)$$

The Lévy scaling prediction $\hat{S}_q(\tau) \propto \tau^{q/\alpha}$ implies the log-log regression slope $\hat{\zeta}(q) = q/\hat{\alpha}_q$. The structure-function estimator $\hat{\alpha}_{\text{sf}}$ is the median of $\hat{\alpha}_q$ across the six moment orders. Fractional orders $q < 1$ downweight large increments and converge reliably on moderate samples, while $q \in (1, \alpha)$ probe the tail more directly. Systematic concavity of $\hat{\zeta}(q)$ would signal multifractal corrections; the present analysis uses the structure functions as a cross-check on the Lévy self-similarity assumption rather than a test for multifractality.

E. Intraday seasonality correction

Equity-index futures exhibit a well-documented U-shaped intraday volatility pattern [22], [23]. The seasonality model decomposes the 1-minute increment as $\delta_k = \hat{\sigma}_{m(k)} \cdot \tilde{\delta}_k$, where $m(k) \in \{1, \dots, M\}$ is the minute-of-session and $\hat{\sigma}_m$ is the deterministic intraday factor. The profile is estimated as the cross-day mean absolute increment at each minute-of-session:

$$\hat{\sigma}_m = \frac{1}{D_m} \sum_{d: m(k)=m} |\delta_k|, \quad (12)$$

where D_m is the number of trading days with an observation at minute m . The profile is normalized to mean 1 across the session. The de-seasonalized increment is $\tilde{\delta}_k = \delta_k / \hat{\sigma}_{m(k)}$, and the

de-seasonalized cumulative price series is $\tilde{P}(t) = \sum_{k=1}^t \tilde{\delta}_k$. The full PDF analysis is rerun on $\Delta\tilde{P}_\tau$, and the resulting $\hat{\alpha}$ is compared to the baseline raw estimate to isolate the effect of the seasonality removal.

F. Robustness analysis

A full-factorial sensitivity analysis spans all combinations of bin resolution $\varepsilon \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, lag grid density $N_\tau \in \{20, 37\}$, preprocessing (raw or de-seasonalized), and estimator (tip, collapse, structure functions). Every combination is run for all eight markets, producing a robustness matrix of $\hat{\alpha}$ estimates. The contribution of each factor to the cross-cell variance of $\hat{\alpha}$ is quantified by a variance decomposition, identifying which methodological choices drive the most estimation uncertainty.

G. Bias correction

Sample autocorrelations $\hat{\rho}(k)$ are estimated from the sequence of 1-minute price increments for lags $k = 1, \dots, K_{\max} = 60$. The variance ratio at each τ is then constructed from the identity in Equation (5):

$$\widehat{VR}(\tau) = 1 + 2 \sum_{k=1}^{\min(\tau-1, K_{\max})} \left(1 - \frac{k}{\tau}\right) \hat{\rho}(k). \quad (13)$$

The dependence correction term is $b(\tau) = -\frac{1}{2} \log \widehat{VR}(\tau)$. The bias-adjusted response is $y^*(\tau) = \log \hat{P}(0, \tau) - b(\tau)$, and the corrected OLS regression of $y^*(\tau)$ on $\log \tau$ gives $\hat{\alpha}_{\text{corr}} = -1/\hat{m}_{\text{corr}}$, recovering Equation (7).

The correction is implemented as a constrained fit with the coefficient of $b(\tau)$ fixed at 1, rather than a free two-regressor regression. On the project lag grid $\log \tau$ and $b(\tau)$ are highly collinear (both are nearly monotone functions of τ), so an unconstrained regression would produce unstable coefficient estimates regardless of whether the underlying theory is correct. The constrained approach bypasses this collinearity by imposing the theoretical coefficient directly. The sensitivity of $\hat{\alpha}_{\text{corr}}$ to the truncation $K_{\max}$ is assessed as a robustness check.

V. RESULTS

A. Baseline scaling

Table IV reports the tip-scaling estimate $\hat{\alpha}_{\text{tip}}$ for all eight markets, averaged over $\varepsilon \in \{0.1, \dots, 0.5\}$ and $N_\tau \in \{20, 37\}$. Seven of the eight markets yield $\hat{\alpha}_{\text{tip}}$ lying within the interval (1.73, 1.84), consistent with the truncated Lévy flight picture of Mantegna and Stanley [3], [5] and with prior estimates for equity-index instruments [6], [8]. The goodness-of-fit measure R^2 for the log-log regression (Equation (9)) exceeds 0.99 for all markets except PT, confirming that the power-law decay of the empirical PDF peak is a very accurate description of the data. The four North American equity-index futures — YM ($\hat{\alpha} \approx 1.76$), ER ($\hat{\alpha} \approx 1.77$), NQ ($\hat{\alpha} \approx 1.75$) and ES ($\hat{\alpha} \approx 1.78$) — cluster tightly near the empirical consensus value of $\alpha \approx 1.7$ –1.8. The three European contracts are slightly more dispersed: CAC-40 (CF) sits at the lower end ($\hat{\alpha} \approx 1.73$),

DAX (DA) at $\hat{\alpha} \approx 1.75$, and FTSE 100 (FT) at the top of the non-PT range ($\hat{\alpha} \approx 1.83$).

a) *PT diagnostic.*: The S&P/TSX 60 futures contract (PT) is a persistent outlier in every analysis carried out in this paper ($\hat{\alpha}_{\text{tip}} \approx 2.6$, $R^2 \approx 0.91$, largest inter-estimator spread). Three features of the PT data suggest that excess price discreteness, rather than a genuinely different tail process, drives the anomaly. First, the σ_1/Δ ratio for PT is 1.11 (Table II), barely above the threshold of 1 and far below the ratios of 2–5 seen in the other seven contracts; a large fraction of 1-minute intervals therefore record a zero price change, producing a probability mass at zero that inflates the empirical PDF peak independently of the Lévy body. Second, the proportion of zero-increment minutes in the PT sample exceeds 40%, compared with below 5% for the most liquid contracts (ES, NQ); these zeros compress the empirical peak values at short lags and distort the log-log regression slope. Third, the tick size for PT (CAD 1 per contract) is large relative to intraday volatility, so the integer-scaled price increments occupy only a small number of distinct values at short lags, violating the continuous-distribution assumption underlying both the peak formula and the collapse estimator. Taken together, these factors suggest that the PT anomaly is a microstructure artifact rather than evidence of a different distributional regime; the contract is retained in the cross-section for completeness but its $\hat{\alpha}$ estimates should not be interpreted on the same footing as those for the seven more liquid markets.

B. Effect of intraday seasonality

Removing the U-shaped intraday volatility profile has a clear but market-specific effect on $\hat{\alpha}_{\text{tip}}$. Table III shows the raw and de-seasonalized tip estimates, together with their difference $\Delta\hat{\alpha}_{\text{seas}} = \hat{\alpha}_{\text{tip,des}} - \hat{\alpha}_{\text{tip,raw}}$. Across the eight markets the shift ranges from -0.002 (ER) to $+0.132$ (PT), with the majority of contracts exhibiting a small positive shift: de-seasonalization raises $\hat{\alpha}$ for DA (+0.060), NQ (+0.094), ES (+0.093) and FT (+0.064), while it lowers $\hat{\alpha}$ slightly for CF (-0.016) and YM (-0.009). The effect on regression quality is negligible: ΔR^2 is below 0.003 in absolute value for every market.

Two implications follow. First, deterministic intraday seasonality is not a dominant driver of the fitted Lévy exponent; the broad Lévy scaling picture survives preprocessing. Second, the direction of the shift is market-specific, indicating that the intraday volatility profile interacts with the distribution shape differently across asset classes. For the remaining analyses the raw (un-normalized) series is used as the primary input, consistent with the standard in the literature.

C. Estimator comparison

Table IV compares $\hat{\alpha}_{\text{tip}}$, $\hat{\alpha}_{\text{sf}}$, and $\hat{\alpha}_{\text{collapse}}$ for all eight markets. The three estimation principles produce broadly consistent but not identical results. Across six of the eight markets, the estimates are systematically ordered: $\hat{\alpha}_{\text{tip}} < \hat{\alpha}_{\text{sf}} < \hat{\alpha}_{\text{collapse}}$. The spread between the smallest and largest

TABLE III
INTRADAY SEASONALITY EFFECT ON $\hat{\alpha}_{\text{tip}}$

Market	$\hat{\alpha}_{\text{tip,raw}}$	$\hat{\alpha}_{\text{tip,des}}$	$\Delta\hat{\alpha}_{\text{seas}}$	ΔR^2
CF	1.731	1.747	+0.016	+0.000
DA	1.750	1.810	+0.060	+0.001
ER	1.771	1.769	-0.002	+0.001
ES	1.784	1.877	+0.093	-0.002
FT	1.834	1.898	+0.064	+0.000
NQ	1.750	1.844	+0.094	-0.002
PT	2.631	2.763	+0.132	+0.001
YM	1.758	1.749	-0.009	+0.000
$\varepsilon \in \{0.1, \dots, 0.5\}$, $N_\tau \in \{20, 37\}$; $\Delta R^2 \equiv R_{\text{des}}^2 - R_{\text{raw}}^2$.				

estimate within a single market is typically 0.15–0.35 exponent units for the non-PT contracts. This systematic ordering is not sampling noise. The tip estimator depends only on the central value of the PDF at each lag; the structure-function estimator weights the entire distribution, with greater emphasis on the tails for larger q ; and the collapse-quality estimator penalizes global misalignment of the rescaled PDFs. The fact that different distributional features select different effective exponents confirms that the data are only *approximately* described by a single self-similar Lévy law.

The structure-function analysis provides additional evidence for approximate rather than exact scaling: the implied exponent $\alpha(q) = q/\zeta(q)$ is not constant across q but increases monotonically with the moment order, consistent with the truncated Lévy flight picture in which exact self-similarity fails in the far tails [5], [6]. For ES (the pilot market), $\alpha(q)$ rises from approximately 1.74 at $q = 0.25$ to 1.96 at $q = 1.50$, bracketing the tip and collapse estimates.

PT is again an outlier: the inter-estimator gap is substantially larger than for the other seven markets ($\hat{\alpha}_{\text{tip}} \approx 2.63$ versus $\hat{\alpha}_{\text{sf}} \approx 1.82$ and $\hat{\alpha}_{\text{collapse}} \approx 2.48$), suggesting that the single-exponent Lévy description is considerably less adequate for that contract.

D. Robustness

The full-factorial robustness matrix enumerates all combinations of resolution $\varepsilon \in \{0.1, \dots, 0.5\}$, lag-grid density $N_\tau \in \{20, 37\}$, preprocessing (raw or de-seasonalized), and estimator (tip, sf, collapse), yielding a comprehensive picture of which design choices drive estimation uncertainty.

Two findings stand out. First, within a fixed estimator, $\hat{\alpha}$ is essentially insensitive to ε and N_τ : the standard deviation of $\hat{\alpha}$ across the five resolution values and two grid densities is below 0.025 for every market and estimator. Second, the estimator choice is the dominant source of variation. For the same market, preprocessing, ε , and N_τ , the mean fitted exponents across estimators differ by 0.1–0.2 units. This inter-estimator gap is comparable to or larger than the inter-market spread among the seven non-PT contracts ($\hat{\alpha}_{\text{tip}}$ ranges from 1.73 to 1.83), which implies that reported $\hat{\alpha}$ values from different studies using different estimators cannot be directly compared without controlling for the estimation principle.

The gap between tip and the other two estimators is also informative about the mechanism studied in Section V-F: the

tip estimator is the most sensitive to the central PDF value at each lag, which is where the short-term mean-reversion bias is concentrated. This observation motivates the bias correction applied in that section.

E. Asymmetry

The symmetric Lévy model assumes that the positive and negative tails of the increment distribution have the same shape. This is tested via the pointwise relative asymmetry ratio $A(x, \tau) = [\hat{P}(x, \tau) - \hat{P}(-x, \tau)] / [\hat{P}(x, \tau) + \hat{P}(-x, \tau)]$, which is zero for a perfectly symmetric distribution and ± 1 at the extremes. The scalar summary per market is the mean of $|A(x, \tau)|$ over $x > 0$, averaged further over the full lag grid.

Table V reports these values across markets. The most symmetric market is ER (0.27), followed by NQ (0.32) and YM (0.35); the least symmetric are PT (0.54) and FT (0.51), with DA (0.45) also above the cross-market average. These aggregate ratios are dominated by the behavior at the very shortest lags and in the far tails, where histogram estimates are sparse and noisy. In the moderate-lag range where the PDF estimates are most reliable, the mean $|A|$ is generally below 0.1 for all markets, indicating near-symmetry in the region that determines the Lévy fit.

The symmetrized collapse — in which the empirical PDF is replaced by $\hat{P}_{\text{sym}}(x, \tau) = [\hat{P}(x, \tau) + \hat{P}(-x, \tau)]/2$ — matches or marginally improves on the raw collapse for all markets. The improvement in collapse score is small, confirming that the asymmetric component of the distribution does not materially affect the self-similar scaling structure. There is mild but detectable asymmetry at short lags in several markets, consistent with bid-ask bounce asymmetry and short-term momentum in futures order flow; however, its magnitude is small enough that the symmetric Lévy approximation remains a reasonable first-order description.

F. Bias correction

Seven of the eight markets exhibit negative first-order autocorrelation of 1-minute returns; ER is the exception with $\hat{\rho}(1) = +0.001$. Table VI reports the estimated first-order autocorrelation $\hat{\rho}(1)$, the long-run variance ratio $\widehat{VR}_\infty = 1 + 2 \sum_{k \geq 1} \hat{\rho}(k)$, the uncorrected tip estimate $\hat{\alpha}_{\text{tip}}$, the bias-corrected estimate $\hat{\alpha}_{\text{corr}}$, and the implied correction $\widehat{\Delta\alpha} = \hat{\alpha}_{\text{tip}} - \hat{\alpha}_{\text{corr}}$. For reference the table is sorted by $\widehat{VR}_\infty$ in ascending order (strongest mean reversion first).

The correction is systematic for seven of the eight markets: every contract except ER corrects downward ($\widehat{\Delta\alpha} > 0$), and the markets with the smallest $\widehat{VR}_\infty$ (strongest net mean reversion) display the largest corrections. PT, which has $\widehat{VR}_\infty = 0.895$ and $\hat{\rho}(1) = -0.020$, shows the largest correction ($\widehat{\Delta\alpha} = 0.921$). Among markets that correct downward, CF with $\widehat{VR}_\infty = 0.974$ and $\hat{\rho}(1) = -0.007$ shows the smallest correction ($\widehat{\Delta\alpha} \approx 0.04$). ER is the sole exception: with $\hat{\rho}(1) = +0.001$, the variance ratio at short lags slightly exceeds 1, so the correction at those lags acts in the opposite direction, yielding a net upward correction of $|\widehat{\Delta\alpha}| = 0.043$. The monotone ordering of $\widehat{\Delta\alpha}$ with $\widehat{VR}_\infty$ holds across the

TABLE IV
THREE-WAY LÉVY EXPONENT ESTIMATES BY MARKET

Market	$\hat{\alpha}_{\text{tip}}$	R^2_{tip}	$\hat{\alpha}_{\text{sf}}$	$\hat{\alpha}_{\text{collapse}}$	Spread
CF	1.731	0.999	1.883	1.995	0.264
DA	1.750	0.996	1.858	2.100	0.350
ER	1.771	0.997	1.891	2.075	0.304
ES	1.784	0.997	1.872	1.980	0.196
FT	1.834	0.998	1.924	2.190	0.356
NQ	1.750	0.998	1.859	1.768	0.109
PT	2.631	0.913	1.824	2.480	0.807
YM	1.758	0.998	1.881	2.030	0.272
$\varepsilon \in \{0.1, \dots, 0.5\}$, $N_\tau \in \{20, 37\}$; Spread $\equiv \max(\hat{\alpha}_{\text{tip}}, \hat{\alpha}_{\text{sf}}, \hat{\alpha}_{\text{collapse}}) - \min(\hat{\alpha}_{\text{tip}}, \hat{\alpha}_{\text{sf}}, \hat{\alpha}_{\text{collapse}})$.					

TABLE V
MEAN ABSOLUTE ASYMMETRY RATIO BY MARKET

Market	$\bar{A} \equiv A(x, \tau) _{x>0, \tau}$
ER	0.267
NQ	0.324
YM	0.346
ES	0.397
CF	0.443
DA	0.450
FT	0.513
PT	0.537
$\varepsilon = 0.1$, $N_\tau = 20$.	

seven mean-reverting markets, constituting a cross-sectional validation of the bias mechanism derived in Section II-C.

The robustness check that excludes the three shortest lags ($\tau \in \{1, 2, 3\}$), which are most contaminated by bid-ask bounce, leaves the ordering unchanged but reduces the magnitudes slightly, consistent with a contribution from microstructure effects at very short horizons. For the most liquid equity-index futures (YM, CF, ER), the correction on the restricted grid is below 0.02 exponent units, placing the corrected estimates squarely within the historical Lévy-finance benchmarks of [3], [6]. The K_{max} sensitivity analysis on the pilot market (ES) confirms that $\hat{\alpha}_{\text{corr}}$ is stable across $K_{\text{max}} \in \{30, 60, 120, 240\}$, with the corrected estimate varying by less than the inter-estimator spread documented in Section V-C.

TABLE VI
VARIANCE-RATIO BIAS CORRECTION BY MARKET

Market	$\hat{\rho}(1)$	$\widehat{VR}_\infty$	$\hat{\alpha}_{\text{tip}}$	$\hat{\alpha}_{\text{corr}}$	$\widehat{\Delta\alpha}$
PT	-0.020	0.895	2.684	1.764	+0.921
DA	-0.008	0.949	1.758	1.623	+0.135
ES	-0.008	0.949	1.791	1.714	+0.077
YM	-0.004	0.951	1.759	1.708	+0.051
FT	-0.009	0.952	1.842	1.783	+0.059
ER	+0.001	0.958	1.784	1.827	-0.043
CF	-0.007	0.974	1.730	1.695	+0.036
NQ	-0.001	0.980	1.737	1.690	+0.046
$\varepsilon = 0.1$, $N_\tau = 20$, $K_{\text{max}} = 60$; sorted by $\widehat{VR}_\infty \uparrow$.					

VI. CONCLUSION

This paper has examined Lévy scaling in short-horizon futures returns across eight equity-index contracts spanning

North America and Europe, using 1-minute OHLCV data from exchange inception through January 2026. Six findings emerge.

(i) Baseline scaling. Seven of the eight markets yield peak-scaling exponents lying within $\hat{\alpha}_{\text{tip}} \in (1.73, 1.84)$, with $R^2 > 0.99$ for the log-log regression in every case. The cross-market mean is $\bar{\alpha} \approx 1.77$, consistent with the Mantegna–Stanley consensus and with prior estimates for equity-index instruments. The S&P/TSX 60 contract (PT) is a persistent outlier ($\hat{\alpha} \approx 2.6$, $R^2 \approx 0.91$), suggesting the single-exponent Lévy description is less adequate for that instrument.

(ii) Intraday seasonality. Removing the U-shaped intraday volatility profile shifts $\hat{\alpha}_{\text{tip}}$ by at most 0.13 exponent units and has a negligible effect on regression quality ($|\Delta R^2| < 0.003$). The direction of the shift is market-specific. Intraday seasonality is therefore not a dominant confound for Lévy scaling estimation at the 1-minute frequency.

(iii) Estimator consistency. Three independent estimators — tip-scaling OLS, collapse-quality minimisation, and fractional-order structure functions — produce broadly consistent but not identical results. The systematic ordering $\hat{\alpha}_{\text{tip}} < \hat{\alpha}_{\text{sf}} < \hat{\alpha}_{\text{collapse}}$ holds for six of the eight markets, with inter-estimator spreads of 0.15–0.35 units. This gap is comparable to or larger than the inter-market spread among the seven non-PT contracts, implying that reported $\hat{\alpha}$ values from different studies cannot be directly compared without controlling for the estimation principle.

(iv) Robustness. Within any fixed estimator, $\hat{\alpha}$ is essentially insensitive to histogram resolution ε and lag-grid density N_τ : the standard deviation across these design choices is below 0.025 for every market. The estimator choice, not the tuning parameters, is the dominant source of variation in reported $\hat{\alpha}$ values.

(v) Symmetry. The empirical PDFs exhibit mild but detectable asymmetry at short lags, with mean absolute asymmetry ratios ranging from 0.27 (ER) to 0.54 (PT) in aggregate. In the moderate-lag region where PDF estimates are most reliable, $|A|$ falls below 0.1 for all markets. Symmetrising the PDFs prior to collapse matching produces at most a marginal improvement in collapse score, confirming that the symmetric Lévy approximation is a reasonable first-order description.

(vi) Bias correction. Seven of the eight markets exhibit negative first-order autocorrelation; ER is the exception with

$\hat{\rho}(1) = +0.001$. Long-run variance ratios span $\widehat{VR}_\infty \in (0.895, 0.980)$. For the seven mean-reverting markets the variance-ratio correction reduces $\hat{\alpha}_{\text{tip}}$ by $\widehat{\Delta\alpha} \in (0.04, 0.92)$, with the correction monotonically ordered by $\widehat{VR}_\infty$ — a cross-sectional validation of the bias mechanism. ER corrects slightly upward ($\widehat{\Delta\alpha} = -0.04$) due to its marginally positive $\hat{\rho}(1)$. The corrected estimates for the most liquid equity-index futures fall within 0.02 exponent units of historical benchmarks.

a) Limitations.: Several caveats apply. The study covers a single asset class; whether the $\alpha \approx 1.7$ –1.8 finding extends to foreign exchange, commodity, or fixed-income futures is an open question. The peak-scaling estimator targets the body of the distribution rather than the tail index in the Hill–Pickands sense, and the two need not coincide for truncated Lévy flights. The bias correction is derived under covariance-stationarity; at the very shortest lags, microstructure effects such as bid-ask bounce and price discreteness can introduce deviations that the correction does not fully absorb.

b) Extensions.: Longer samples would allow the temporal stability of $\hat{\alpha}$ to be assessed and would permit the cross-market correlation of exponent estimates to be studied as a possible systemic risk indicator. Applying the same framework to GARCH-filtered residuals would separate the contribution of conditional heteroscedasticity from the unconditional tail behaviour. Finally, extending the lag grid to sub-minute frequencies, where Lévy scaling is expected to break down due to price-grid discreteness, would sharpen the lower boundary of the self-similar regime.

ACKNOWLEDGMENT

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